Advising Assistant Project Stage 5:

Team accomplishments:

During this stage, our team focused on systematically testing the Advising Assistant system using both acceptance testing (use case testing) and unit testing (white-box and black-box methods).

We executed multiple test runs for each use case, covering both basic flows and alternative flows, and verified that the system behaves as expected.

We also performed:

- White-box testing with 100% branch coverage
- Black-box testing using equivalence partitioning
- Debugging and fixing errors found during testing

Team Member	Tasks
Isabella	Performed acceptance testing for login, registration, transcript
Isabella	Tested recommendation engine and report generation
Ingrid	Developed white-box tests and control flow graph
Ingrid	Developed black-box tests and documented results

Acceptance Testing:

Use Case: Generate Recommendation

Test Run 1: Basic Flow

Input:

- User: student1
- Transcript/ courses taken: CYBI1, MATH1

Steps:

1. Login
2. Click “Generate Recommendation”

Expected Output:

- CYBI2, MATH2

Actual Output:

- CYBI2, MATH2

Result: Pass

Test Run 2: Alternative Flow

Input:

- Completed: all courses

Steps:

1. Login
2. Click “Generate Recommendation”

Expected Output:

- "No remaining required courses”

Actual Output:

- Empty list displayed

Result: Pass

Use Case: View Transcript

Test Run 1: Basic Flow

Input:

- User: student1
- Courses taken: CYBI1, MATH1

Steps:

1. Login
2. Click “Transcript”

Expected Output:

- CYBI1, MATH1

Actual Output:

- CYBI1, MATH1

Result: Pass

Test Run 2: Alternative Flow

Input:

- Invalid student

Steps:

1. Login
3. Click “Transcript”

Expected Output:

- Error message

Actual Output:

- "Student not found”

Result: Pass

Use Case: Login

Test Run 1: Basic Flow/ Valid Login

Input:

- Valid/ existing username
- Valid/ existing password

Steps:

1. Click “Login”

Expected Output:

- Login, dashboard

Actual Output:

- Login, dashboard

Result: Pass

Test Run 2: Alternative Flow

Input:

- Invalid login credentials

Steps:

1. Click “Login”

Expected Output:

- Error/ stay on login

Actual Output:

- Reload login page

Result: Pass

Use Case: No Completed Courses/ Missing Prerequisites

Test Run 1: Basic Flow

Input:

- Completed: []
- Degree Plan: CYBI1, CYBI2
- Prerequisites: CYBI2: {CYBI1}

Steps:

1. Login
2. Click “Generate Recommendations”

Expected Output:

- CYBI1

Actual Output:

- CYBI1

Result: Pass

Test Run 2: Alternative Flow

Input:

- Completed: all courses

Steps:

3. Login
4. Click “Generate Recommendation”

Expected Output:

- "No remaining required courses”

Actual Output:

- Empty list displayed

Result: Pass

Use Case: User Registration

Test Run 1: Basic Flow

Input:

- Username: newstudent
- Password: 123
- Role: student

Steps:

1. Navigate to Register page
2. Enter credentials
3. Submit form

Expected Output:

- Account created
- Redirect to login

Actual Output:

- Account created successfully
- Redirected to login page

Result: Pass

Test Run 2: Alternative Flow: Duplicate Username

Input:

- Username: student1

Expected Output:

- Error message: “Username already exists”

Actual Output:

- Error message displayed

Result: Pass

White-Box Testing: Recommendation Engine

generation_recommendation()

Use Case: Logout

Test Run 1: Basic Flow

Steps:

1. User clicks Logout

Expected Output:

- Session cleared
- Redirect to login

Actual Output:

- Redirected to login page

Result: Pass

Test Run 2: Alternative Flow: Access after logout

Steps:

1. Logout
2. Try /dashboard

Expected Output:

- Redirect to login

Actual Output:

- Redirected correctly

Result: Pass

Use Case: View Transcript

Test Run 1: Basic Flow

Input:

- Logged in as student1

Steps:

1. Click “View Transcript”

Expected Output:

- Transcript displayed

Actual Output:

- Courses displayed correctly

Result: Pass

Test Run 2: Alternative flow: unauthorized access

Steps:

1. Login as student1
2. Navigate to /transcript/student2

Expected Output:

- Access denied

Actual Output:

- “Access Denied” displayed

Result: Pass

Use Case: View Degree

Expected Output:

- Shows completed + remaining courses

Actual Output:

- Correct courses displayed

Result: Pass

Test Run 2: Alternative Flow (Missing Data)

Expected Output:

- Error message

Actual Output:

- Handled or displayed safely

Result: Pass

Use Case: View Advisees (Faculty)

Test Run 1: Basic Flow

Input:

- Login as faculty1

Expected Output:

- List of students displayed

Actual Output:

- student1 shown

Result: Pass

Test Run 2: Alternative Flow (No Advisees)

Expected Output:

- “No advisees” message

Actual Output:

- Empty list or message

Result: Pass

Use Case: Generate Course Recommendation

Test Run 1: Basic Flow

Input:

- Completed: CYBI1, MATH1

Expected Output:

- CYBI2, MATH2

Actual Output:

- CYBI2, MATH2

Result: Pass

Test Run 2: Alternative Flow (all completed)

Expected Output:

- No recommendations

Actual Output:

- Empty list shown

Result: Pass

Use Case 8: Generate Report

Test Run 1: Basic Flow

Steps:

1. Click “Generate Report”

Expected Output:

- Report with:
    - Completed courses
    - Remaining courses
    - Recommendations

Actual Output:

- Report displayed correctly

Result: Pass

Test Run 2: Alternative Flow (Invalid Student)

Expected Output:

- Error message

Actual Output:

- “Student not found”

Result: Pass

Use Case 9: Course Detail Page

Test Run 1: Basic Flow

Steps:

1. Click a course link

Expected Output:

- Course description + credits

Actual Output:

- Correct info displayed

Result: Pass

Test Run 2: Alternative Flow (invalid course)

Steps:

- Navigate to /course/XYZ

Expected Output:

- Error message

Actual Output:

- “Course not found”

Result: Pass

Use Case 10: Role-based access

Test Run 1: Basic Flow

Input:

- Login as faculty

Expected Output:

- See advisees

Actual Output:

- Advisee list displayed

Result: Pass

Test Run 2: Alternative Flow (student restricted access)

Steps:

- Student tries to access another student

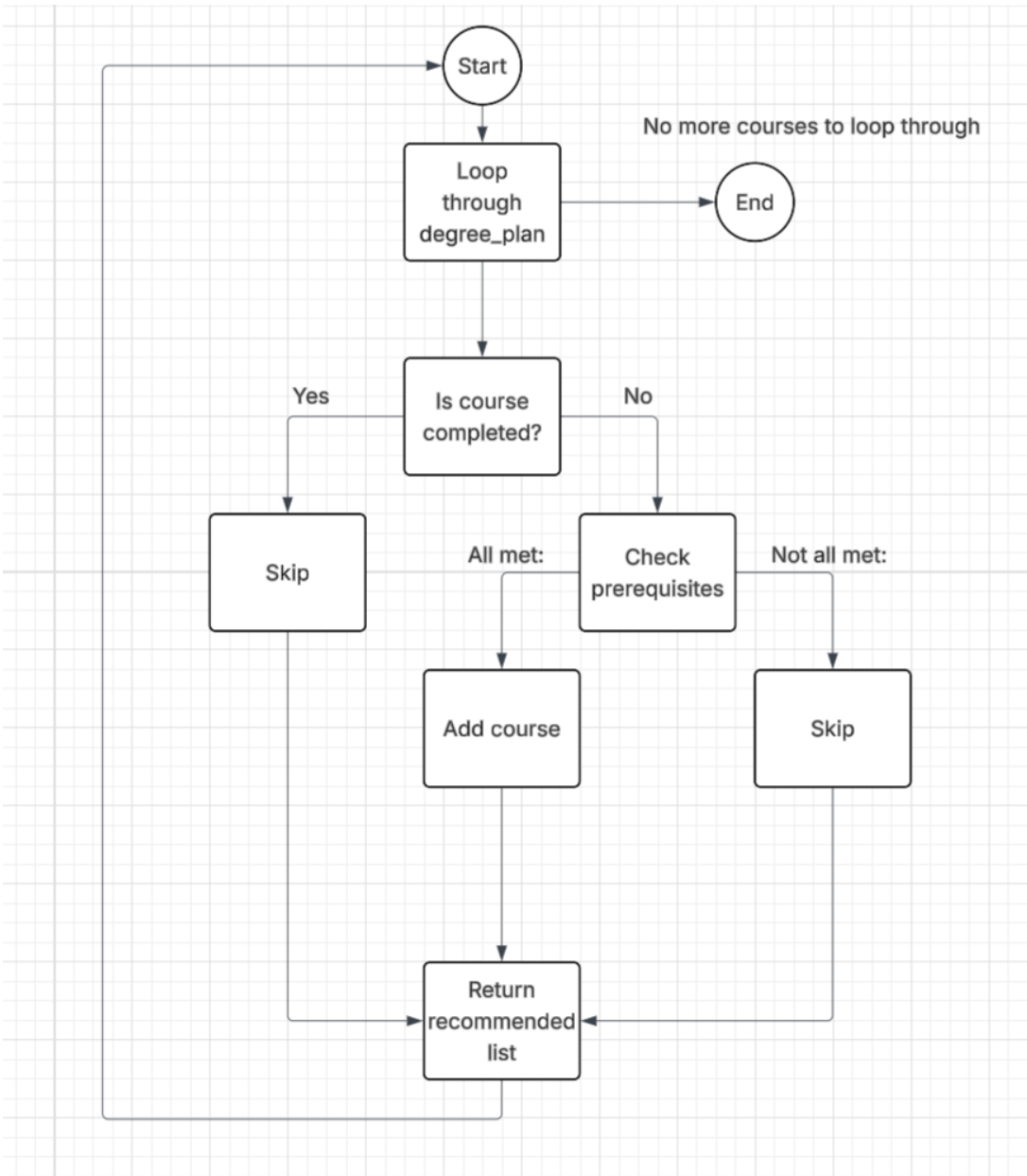
Expected Output:

- Access denied

Actual Output:

- Access denied

Result: Pass



Branches:

1. Course already completed
2. Course not completed + prereqs met
3. Course not completed + prereqs not met

Test ID	Input	Covered Path	Expected	Actual	Notes
1	completed=["CYBI1"]	prereq met	["CYBI2"]	["CYBI2"]	PASS
2	completed=[]	prereq NOT met	["CYBI1"]	["CYBI1"]	PASS

3	completed=["CYBI1","CYBI2"]	already completed	[]	[]	PASS

Black-Box Testing:

generate_recommendations()

Equivalence Partitioning

Class	Description
ECI	Valid input w/ partial completion
EC2	No completed courses
EC3	All courses completed
EC4	Invalid input (missing prerequisites data)

Test Table:

Test ID	Input	Equivalence Class	Expected	Actual	Notes
BB1	["CYBI1"]	EC1	["CYBI2"]	["CYBI2"]	PASS
BB2	[]	EC2	["CYBI1"]	["CYBI1"]	PASS
BB3	all done	EC3	[]	[]	PASS
BB4	missing prereqs	EC4	skip prereq logic	worked	handled safely

Error handling:

Issue:

Course detail page caused NameError: courses not defined

Fix: Imported courses from data.py

Another issue:

Recommendation engine failed when prerequisites missing

Fix: Added this line oof code:

Prereqs = prerequisites.get(course, [])

Team Members:	Completion of the	Team Engagement (0-5)	Justifications for point deductions

	assigned task (0-20)		
Ingrid Alvarez	20	5	N/A
Isabella Morgan	20	5	N/A